MIL-V-62475(AT)
24 June 1985
SUPERSEDING
(see 6.7)

MILITARY SPECIFICATION

VEHICLE, FIRE SUPPORT TEAM, M981
(FISTV)

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a full-tracked, land combat, fire support team vehicle, with a high degree of tactical maneuverability.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical.

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8604 - Welding, Fusion; Aluminum Alloys; Process and Performance of.
MIL-W-8611 - Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for.
MIL-C-22750 - Coating, Epoxy Polyamide.
MIL-P-23377 - Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-P-27401 - Propellant Pressurizing Agent, Nitrogen.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-H-46170 - Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.
MIL-P-52192 - Primer Coating, Epoxy.
MIL-C-62310 - Carriers, Armored, Full-Tracked: M113A2, M113A2/EFT, M106A2, M125A2, M557A2 and M981, Chassis, Weapon, Full-Tracked: M7541A1 and M901.
MIL-T-62327 - TOW Vehicle: Combat, Anti-Tank (Less Tow Weapon), M901, M901A1 and M981 Fire Support Team Vehicle; Processing for Storage and Shipment of.
MIL-R-62366 - Ring, Electrical Contact, M901 and M981.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

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MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1246	- Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1275	- Characteristics of, 28 V dc Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.
MS 35000	- Battery, Storage, Lead-Acid, Waterproof.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

8750031	- Carrier, Personnel, Full Tracked, Armored, Fire Support, M981.
12306657	- Transportability Data, APC M981.

PURCHASE DESCRIPTIONS

M981PD-S-62476	- M981 Targeting Station.
M981PD-C-62478	- Optical Coupler.
M981PD-C-62479	- Targeting Station Control and Display.
M981PD-G-62480	- North Seeking Gyrocompass Function Specification.
M981PD-I-62481	- Control, Intercommunication Unit.
M981PD-G-62487	- North Seeking Gyrocompass Fabrication Specification.
M981PD-G-62488	- Amplifier, Audio Frequency.

Detailed Test Plan, Nuclear Effects Subtest of Development Test II (PQT-G) of FISTV, Annex A, October 1982 (confidential).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish vehicle(s) which shall be subjected to first article inspection (see 4.4.1 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent vehicles delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.6.1 and 6.5).

3.2.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings (see 4.6.1).

3.2.2 Radioactive materials. Use of radioactive materials shall be prohibited, except for the night sight. The night sight lens coating contains thorium flouride. The only potential hazard is from inhaling or swallowing this material. Broken lenses shall be disposed of in accordance with applicable Army Regulation (see 4.6.1 and 6.6.1).

3.2.3 Hydraulic fluid. The hydraulic fluid used shall be in accordance with MIL-H-5606, MIL-H-6083 or MIL-H-46170. The system performance will be degraded when MIL-H-46170 is used at temperatures of 0°F or below (see 4.6.1).

3.2.3.1 Hydraulic fluid cleanliness. The hydraulic fluid system shall conform to cleanliness level 300 as specified in MIL-STD-1246 (see 4.6.1).

3.3 Design and construction. Vehicles, components, subassemblies and assemblies shall be manufactured and assembled into a complete vehicle in accordance with Drawing 8750031 and applicable specifications and documents. Vehicles shall meet all requirements specified herein and be of a quality that assures a vehicle free from defects that would compromise, limit, or reduce its performance (see 4.6.1).

3.3.1 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products List (QPL's) whenever available. Contractor inspection records shall specifically list all QPL items by number and date of the QPL, name of supplier, and part or drawing number(s). When parts and assemblies are approved as qualified products but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of supplier(s) (see 4.6.1).

3.3.2 Batteries. The FISTV vehicle power source shall consist of four lead-acid batteries 6TN (MS 35000 or equivalent) connected in a series-parallel configuration to provide 24 volts (V) direct current (dc) (see 4.6.1).

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3.3.2.1 Battery specific gravity. All four batteries shall show no presence of battery acid around the battery case or on top of batteries. The electrolyte level shall cover the plates and be just below the level indicator. The specific gravity level of each cell shall be at least 1.250 at 80 degrees Fahrenheit (°F) [26.7 degrees Celsius (°C)] (see 4.6.1).

3.3.3 Alternator. With the engine running at 2165 ± l00 revolutions per minute (rpm), the voltage regulator rheostat shall be adjusted to assure that the FISTV vehicle alternator will generate 28 ± 0.5 V dc and 200 ± 5 amperes (A) (see 4.6.1).

3.3.4 Electrical connectors. Electrical connectors shall be in accordance with MIL-STD-454, requirement 10. The soldering of connectors shall be in accordance with MIL-STD-454, requirement 5. When brazing is used, it shall be in accordance with MIL-B-7883 (see 4.6.1).

3.3.5 Wiring harness. Wiring harnesses and cables shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.1).

3.3.6 Nitrogen.

3.3.6.1 Image transfer assembly (ITA). The pressurized sections of the ITA shall be initially purged and charged with nitrogen, conforming to BB-N-411, type I, class 1, or grade A or B (see 4.6.1).

3.3.6.2 Accumulator precharge. The accumulator shall be pressurized to the precharge using nitrogen gas specified by MIL-P-27401, type I, grades A, B or C or BB-N-411, type I, grades A or B.

3.4 Physical characteristics.

3.4.1 Weight. The FISW (M98l) vehicle weight, when combat loaded in accordance with applicable drawings, shall be approximately 26 900 pounds (lbs) [12 200 kilograms (kg)] (see 4.6.3.1).

3.4.2 Basic issue items (BII). The FISTV vehicle shall be capable of storing all BII and Government Furnished Equipment (GFE). Items shall fit properly and shall not interfere with the operation of the vehicle and components in any manner (see 4.6.3.2).

3.4.3 Component removal. The removal of all FISTV vehicle components specified for dismount operation (see table I) to a distance of 50 ± 5 meters (m) [165 ± 15 feet (ft)] from the vehicle, shall require not more than five minutes with a four-man crew. Remounting of the same equipment with a four-man crew shall require not more than ten minutes, including boresight collimation time (see 4.6.3.3).

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TABLE I. Dismounted operable FISTV components.

ITEM	
1.	AN/TVQ-2, G/VLLD (including items for ground operation)
2.	AN/TAS-4B, Closed cycle cooler night sight (including collimator and battery case)
3.	AN/PSG-2A or AN/PSG-5 DMD (including cables and battery)
4.	AN/GRC-160, Radio (including antenna battery)
5.	AN/GRA-39, Radio set control group (remote)

3.4.4 Transportability. The FISTV vehicle shall be capable of being transported in the existing C-130 aircraft in accordance with Drawing 12306657. In this air transport configuration, the vehicle shall not exceed a vertical height of 103.5 inches (in) [2628 millimeters (mm)] when the chassis deck is not higher than 73 in (1854 mm) (see 4.6.3.4).

3.4.5 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications or welding procedures which have been approved by the procuring activity. Welding of aluminum alloys shall be in accordance with MIL-W-8604. Welding of steel shall be in accordance with MIL-W-8611 (see 4.6.3.5).

3.5 Performance. A complete FISTV vehicle, loaded to approximately 26 900 lbs (12 200 kg) shall perform as specified herein with the targeting head in the stowed position unless otherwise noted. The FISTV, serviced and equipped for prevailing climatic conditions, shall operate as specified with or without special equipment. Performance as specified herein, shall be demonstrated on smooth, level, hard-surfaced roads free of loose material. A section of the road shall have an artificial bump as shown in figure 1 and it shall be traversed at least once every five miles [8 kilometers (km)] (see 4.6.4).

3.5.1 FISTV system.

3.5.1.1 Reliability. The FISTV vehicle system shall have a minimum acceptable value mission reliability of 0.8 when operated in accordance with the composite mission profile (see 3.5.3). This value is based on the GFE reliability as shown in table II (see 4.6.4.1).

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3.5.1.2 Maintainability.

3.5.1.2.1 Maximum time to repair (MAXTTR). The maximum time to repair for the FISTV system, excluding the chassis and optics, shall be 0.7 hours. This time must include 90 percent of all corrective maintenance actions (see 4.6.4.1).

3.5.1.2.2 Maintenance ratio (MR). The FISTV vehicle system shall have an MR in terms of maintenance manhours (MMH) divided by operating hours that does not exceed the values stated for the following categories of maintenance (see 4.6.4.1):

	Automotive	FISTV Unique Equipment	Total System
Organization	0.43	0.07	0.50
Direct Support/General Support	0.11	0.12	0.23

TABLE II. GFE reliability.

Item	Reliability Baseline
Vehicle	0.9450
Night sight	0.9987
TSEC/KY-57 COMSEC	0.9960
Digital message device	0.9865
Remoted G/VLLD	0.9652
PLRS	0.9987
Radios	0.9996
Other GFE	1.0000

NOTE: The above results in a mission reliability baseline of 0.8935 for the FISTV vehicle GFE items.

3.5.2 Automotive operation. The FISTV vehicle shall meet all the operational requirements regarding reliability, durability and maintainability as specified in MIL-C-62310 for the modified M113A2 chassis, except where otherwise specified herein (see 4.6.4.1).

3.5.3 Mission profile. The FISTV vehicle mission profile shall consist of two mission profiles. A total of 54 miles (87 km) shall be accumulated on primary roadways at 18-25 miles per hour (mph) [14.5 kilometers per hour (km/hr)] in nine mile (14.5 km) increments. This profile shall consist of one operational scenario, 15 ranging scenarios and five designating scenarios per segment with three segments per profile (see 4.6.4.2.1).

3.5.4 Speed. The FISTV vehicle shall sustain speeds of 32 mph (51 km/hr). The targeting station shall be in the stowed position for norma1 travel. Head erect travel shall not exceed 50 m (164 ft) with the speed not to exceed 5 mph (8 km/hr) (see 4.6.4.3).

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3.5.5 Braking. The FISTV vehicle shall stop within a distance of 40 ft (12.2 m) from the point of service brake application when traveling at 20 mph (32 km/hr). The drift within the stopping distance shall not exceed 3 ft (0.91 m). Steering action by the driver is permitted to minimize drift during braking (see 4.6.4.4).

3.5.6 Holding. The FISTV vehicle shall hold stationary when on a 60 percent longitudinal grade with the service brakes applied. On the same grade, the FISTV vehicle shall not move with the brakes locked for parking and the transmission in neutral (see 4.6.4.5).

3.5.7 Shallow water crossing. The FISTV vehicle shall be capable of fording shallow hard-bottomed, salt or fresh water channels up to the depth required to float the vehicle. Water leakage into the hull shall not exceed one quart [0.95 liter (L)] per minute (see 4.6.4.6).

3.5.8 Amphibious operation. The FISTV vehicle shall travel at speeds of up to 3 mph (5 km/hr) when afloat in calm water not less than 15 ft (4.6 m) deep. The vehicle shall be capable of operating within a 30 ft (9.1 m) corridor when crossing calm, open water. Water leakage into the hull shall not exceed one quart (0.95 L) per minute (see 4.6.4.7).

3.5.9 Sealing.

3.5.9.1 Leakage. The FISTV vehicle, with all hatches closed and all the components properly installed, shall restrict the leakage of water into the vehicle to not more than 10 drops per minute (see 4.6.5.8.1 and 6.4.1).

3.5.9.2 Bearing seals. Bearing seals shall prevent the entrance of foreign matter into bearings which are exposed to contamination during amphibious or fording operation or when operating in mud, sand, or snow. Seals shall restrict lubricant leakage. Seals designed to allow exit of flushing lubricant may evidence such leakage (see 4.6.5.8.2).

3.5.10 Trench crossing. The FISTV vehicle shall cross trenches up to 36 in (914 mm) in depth and up to 66 in (1676 mm) in width without damage to the chassis (see 4.6.4.9).

3.5.11 Vertical obstacles. The FISTV vehicle shall negotiate vertical obstacles up to 24 in (610 mm) high at low speeds and in forward gear. In reverse gear, the FISTV vehicle shall cross over vertical obstacles up to 14 in (356 mm) high without damage to the chassis (see 4.6.4.10).

3.5.12 Climbing. The FISTV vehicle shall ascend a 60 percent slope in forward and reverse gears from a standstill position at the base of the slope without stalling or damaging the power plant or power train. The FISTV vehicle shall traverse a 30 percent left and right lateral slope in forward and reverse gear (see 4.6.4.11).

3.5.13 Acceleration. The FISTV vehicle shall accelerate from 0 to 20 mph (32 km/hr) in a maximum of 13 seconds on a smooth, level, hard-surfaced road from a standstill position with the transmission in the 1-3 range and the engine at idle speed (see 4.6.4.12).

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3.5.14 Hatches. The driver's hatch shall provide three positions; closed, partially open and fully open. The targeting station and cargo hatch shall open and close. Positive locks shall be provided for all hatches in all required positions (see 4.6.4.13).

3.5.14.1 Hatch transition time. The transition time from open hatch (travel mode) and closed hatch (tactical mode) and vice versa shall require not more than 20 seconds (see 4.6.4.13.1).

3.5.15 Ramp system. The ramp raising time, from a fully lowered position to a closed position, shall be not more than l8 seconds with the engine operating at not more than 1200 rpm (see 4.6.4.14).

3.5.16 Bilge pumps. The bilge pumps, shall remove water from the vehicle at a rate of at least 44 gallons per minute (gpm) [166.56 liters per minute (L/min)] each, with a discharge head of 4.5 ft [13.45 kilopascals (kPa)] and with the pump motor operating on 27.5 V impressed at the motor (see 4.6.4.15).

3.5.17 Trim vane. The FISTV vehicle trim vane shall properly engage in the extended, retracted and released positions without binding or excessive looseness (see 4.6.4.16).

3.5.18 Turning radius. The FISTV vehicle with the transmission control in differential steer, shall make continuous, full 360 degree turns to the right and to the left, with the rear point of the outer edge of the chassis at all times in contact with or within a 50 ft (15.2 m) diameter circle. When in pivot steer, the chassis shall pivot within a 28.0 ft (8.5 m) diameter circle. The applied force to the steering lever shall not exceed 100 lb [445 newtons (N)] to pivot the chassis (see 4.6.4.17).

3.5.19 Optics.

3.5.19.1 Vision devices. Vision device receptacles incorporated in the vehicle shall accept vision devices defined herein without binding or interference, and provide adequate support during all vehicle operating conditions. All M17 and M26 periscopes and all vision blocks shall be equipped with blackout covers for night operation (see 4.6.4.18.1).

3.5.19.2 Panoramic telescope. The panoramic telescope shall allow the following motion.

Azimuth	± 360 degrees
Elevation	+ 10 degrees
Depression	– 20 degrees

The panoramic telescope shall contain a mil graduated reticle, a laser safety filter, and shall provide a clear optical picture (see 4.6.4.18.2).

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3.5.19.3 Night vision. The FISTV vehicle shall utilize the AN/TAS-4B closed cycle cooler night sight, which shall have compatibility and tracking capability with the Ground-Vehicle Laser Locator and Designator (G/VLLD) for both mounted and dismounted operations. An M19 infrared periscope shall be provided at the driver's station to allow the driver to see while driving at night with the service drive lights blacked out (see 4.6.4.18.3).

3.5.19.3.1 Night sight controls. Night sight controls shall be provided in the targeting station. The night sight contrast, brightness and focus controls shall operate smoothly without binding. The FOV control shall provide smooth switching between the narrow field-of-view (NFOV) and the wide field-of-view (WFOV) without binding, breaking or changes in torque (see 4.6.4.18.3.1).

3.5.19.4 ITA. The ITA shall serve as a unity (1X) periscope between the targeting station, night sight and G/VLLD (through the optical coupler). The ITA shall provide a day surveillance and detection capability with a 25 degree WFOV telescope along a line-of-sight parallel to the G/VLLD in the targeting station (see 4.6.4.18.4).

3.5.19.4.1 Sight selection. The ITA shall provide for selection of an image at the targeting station eyepiece from 3X WFOV, 13X NFOV and Night. The ITA shall require no more than five seconds to switch between any two sight selections (see 4.6.4.18.4.1).

3.5.19.4.2 Image quality.

3.5.19.4.2.1 Image movement. The image transmitted to the targeting station through the ITA shall not exhibit any random movement during targeting station operations which would impair or degrade vision or system tracking performance (see 4.6.4.18.4.2.1).

3.5.19.4.2.2 Internal atmosphere. The ITA, after being pressure sealed, shall exhibit no evidence of moisture, internal condensation, or internal image fogging (see 4.6.4.18.4.2.2).

3.5.19.5 Boresight. The AN/TAS-4B closed cycle cooler night sight shall be mounted such that it can be collimated to the G/VLLD within 0.15 milliradian on a readily repeatable basis. The optical coupler and ITA shall be pupil matched (see 4.6.4.18.5).

3.5.20 Targeting station. The targeting station shall provide for target optical detection, identification, angle determination and target engagement in day or night lighting conditions. Engagement shall consist of directing and adjusting artillery and mortar fire or of designating targets for laser guided munitions (see 4.6.4.19).

3.5.20.1 Azimuth slew mode. The targeting station shall be capable of azimuth traverse over a 360 degree angle at a maximum rate of at least 35 degrees per second (slew mode). This traversing motion shall be controlled by the hand control (see 4.6.4.19.1).

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3.5.20.1.1 Azimuth track mode. The target station shall track at rates down to 0.1 milliradians per second or less. This traversing motion shall be controlled by the hand control (see 4.6.4.19.1.1).

3.5.20.2 Elevation (maximum rate). The elevation time of the targeting head from -23 to +37 degrees shall take no more than five seconds, providing a rate of at least 12 degrees per second (see 4.6.4.19.2).

3.5.20.2.1 Elevation (minimum rate). The targeting head shall elevate and depress at rates down to 0.1 milliradians per second or less (see 4.6.4.19.2.1).

3.5.20.2.2 Azimuth and elevation positional capability. The targeting head shall depress to -23 degrees and elevate to +37 degrees (with respect to the vehicle deck). The targeting head shall rotate 360 degrees minimum to the right and left (see 4.6.4.19.2.2).

3.5.20.3 Pointing and tracking accuracy. The targeting station shall enable operators possessing skills equivalent to that of a qualified G/VLLD operator to achieve tracking accuracies specified below over a period of not less than 20 seconds. The targeting station tracking error shall not exceed G/VLLD tracking error by more than 0.015 milliradians (two sigma error) for either the actual tripod-mounted G/VLLD or the G/VLLD system specification value, whichever is greater. This applies to the tracking of constant velocity targets at ranges from 2-3 km while the FISTV is in a canted position of up to 10 percent in pitch and roll. For ranges of 1-2 km, tracking error shall not exceed G/VLLD tracking error by more than 0.030 milliradians (two sigma error) for either the actual tripod-mounted G/VLLD or the G/VLLD system specification value, whichever is greater (see 4.6.4.19.3).

3.5.20.4 Target locating accuracy. The targeting station subsystem shall be capable of locating targets with 40 m circular error of probability (CEP) accuracy at a range of 3 km from the FISTV position (see 4.6.4.19.3).

3.5.20.5 Position retention. The targeting station mechanism shall retain the position it held (azimuth and elevation) within $\pm$ 5 degrees when electrical power is removed for five minutes, regardless of mounting plane orientation (see 4.6.4.19.4).

3.5.20.6 Slope operation. The targeting station shall operate on 60 percent forward and reverse slopes and on 30 percent side slopes. The targeting station brakes shall prevent any azimuth rotation under all noted slope conditions (see 4.6.4.19.5).

3.5.20.7 Erection drive. The erection drive system shall provide the capability to drive the targeting head and erection arm assemblies to a selected position. The allowable position changes shall be from stow to erect and from erect to stow (see 4.6.4.19.6).

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3.5.20.7.1 Erect position. When the erect mode is selected and the targeting head and erection arm are driven to the erect position, the targeting head shall decelerate prior to achieving the erect position. The erection arm and the elevation drive assembly shall be secured in the erect position by the extension of lock cylinders and the HEAD ERECT indicator lamp shall illuminate (see 4.6.4.19.6.1).

3.5.20.7.2 Erection lock cylinders. When the targeting head is erected, the erection lock cylinders shall engage. The lower erection lock cylinders shall remain engaged, securing the erection arm in the erect position, when hydraulic system pressure has first been bled to the precharge level and then power to the targeting station removed. The erection lock cylinders shall always retract when stow position is selected and hydraulic system pressure is 1200 psi (8.27 MPa) or greater (see 4.6.4.19.6.2).

3.5.20.7.3 Manual erect. The targeting station shall provide a manual stow-to-erect capability, using a hand crank which is inserted into the erection arm. The maximum torque required to erect the targeting head shall not exceed 20 pound-feet (lb-ft) (27.1 newton meters (N m). Engagement of the hand crank shall inhibit the operation of the erection drive motor (see 4.6.4.19.6.3).

3.5.20.7.4 Erection motor speed and time. The erection motor speed shall be controlled so as to limit acceleration and shock loading of the targeting station. The motor shall stop, without hunting or overshooting, when the erection arm reaches the desired position. The time required to erect and lock the targeting head from the stow position shall not exceed eight seconds. The peak erection current shall not exceed 900 A and the steady state erection current shall not exceed 200 A (see 4.6.4.19.6.4).

3.5.20.7.5 Power. Operation of the erection system in any mode shall be prohibited in the absence of both vehicle and equipment power. The erection system shall be driven by the erection drive motor when sufficient power is available. The erection system shall be driven between stow and erect with the hand crank drive on the erection arm, when there is insufficient power available to operate the erection drive motor (see 4.6.4.19.6.5).

3.5.20.7.6 Stow mode. The stow mode shall be provided as the normal travel position. The targeting head and erection arm assemblies shall rest on the stow vertical supports and within the lateral restraints, during all vehicle operating conditions without damage to the vehicle. When the targeting bead is in stow position, all stow indicators shall be illuminated in accordance with table III (see 4.6.4.19.6.6).

3.5.21 Targeting station control and display assembly (TSCD).

3.5.21.1 TSCD display. The TSCD display shall provide a single row of up to twelve alphanumeric characters. The display shall show angles in military mils (least significant digit (LSD) = 1 mil) and distances and positions in meters (LSD = 1 meter except as noted). Alphanumeric characters shall be used to prompt the TSCD operator and provide error messages as necessary in the performance of TSCD related functions (see table IV). When the TSCD power switch is turned on, the display shall flash STANDBY no more than six times and then display V E ###### (see 4.6.4.20.1).

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3.5.21.2 TSCD indicators. The front panel of the TSCD shall provide illuminating indicators. The targeting head status indicators shall prompt the operator as to the status of the targeting head. Interrupt indicators shall be provided to inform of interrupted turret motion and laser fire. Fault indicators shall alert the operator that either built-in-test (BIT) has detected a fault in the TSCD, North Seeking Gyrocompass (NSG) or G/VLLD or their interfaces, or that the vehicle or equipment battery is low. NSG and G/VLLD indicators shall display the status of their respective equipment (see table III) (see 4.6.4.20.2).

3.5.21.3 System test. The test key shall provide a menu to initiate the system-test, examine the current and accumulated BIT status error codes, clear any present faults and observe the NSG mode and self-test status. The TSCD shall display BIT status by means of three alphanumeric error codes; a TSCD error code, a G/VLLD error code and a NSG error code (see 4.6.4.20.3).

TABLE III. TSCD indicator lamps.

Indicator	Color	Meaning
HEAD ERECT	White	Targeting head is full erect.
HEAD READY	Green	Targeting head is erect and G/VLLD cradle locks are in place.
EL ABOVE	Yellow	Head is too high for stow position.
EL BELOW	Yellow	Head is too low for stow position.
EL STOW	White	Head is at proper elevation angle or position to stow.
AZ STOW	White	Head is at proper azimuth angle or position to stow. (179.5° Minimum, 180.5° Maximum)
HEAD STOW	White	Head is in stowed position.
LASER INTRPT	Yellow	Indicates a condition that will prevent laser fire.
CARGO HATCH	Yellow	Indicates the cargo hatch is not securely closed.
DRIVER HATCH	Yellow	Indicates the drivers hatch is not securely closed.
WPN HATCH	Yellow	Indicates the weapon (targeting station) hatch is not securely closed.
NSG READY	Green	NSG is in directional-gyro mode.
NSG ON	Green	NSG power switch is on.
NSG ALIGN	Yellow	Steady – NSG is in the initialization, or realignment mode. Flashing-Vehicle coordinates need to be entered into VEH GRID memory.
GLLD ON	Green	The GLLD rotary function switch is set at RNG 2 (range 2), RNG 1 (range 1) or DES (designate).
NSG FAULT	Red	The TSCD BIT test or the NSG internal BIT has detected an NSG fault.

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TABLE III. TSCD indicator lamps – Continued.

Indicator	Color	Meaning
VEH BAT	Red	Vehicle battery voltage has fallen below 17.5 ± 0.5 V for at least 500 milliseconds (ms).
EQP BAT	Red	The equipment battery voltage has fallen below 20.5 ± 0.5 V for more than 500 ms.
TSCD FAULT	Red	The TSCD's BIT has detected a TSCD fault.
GLLD FAULT	Red	Steady – The TSCD's BIT or G/VLLD internal BIT has detected a GLLD fault. Flashing – G/VLLD power is on but the GLLD code needs to be entered.

TABLE IV. TSCD operator error messages.

Display	Cause
? E>878 KM	Easting value entered was greater than 878 km.
? E<122 KM	Easting value entered was less than 122 km.
N OUT OF RANGE	Northing value entered was greater than 9 999 999 in absolute value.
? ALT>7500 M	Altitude value entered was greater than 7500 meters.
? ALT<-400 M	Altitude value entered was less than –400 meters.
AZ>6399 MIL	Azimuth value entered was greater than 6399 mils.
AZ<0 MIL	Azimuth value entered as negative.
RNG<400 M	Range value entered was less than 400 meters.
OUT OF RANGE	Elevation value entered is greater than 999 mils.
? GLLD CODE	GLLD code number entered contained a 0 or 9, or had less than 3 digits.
NO VALID E/N	An update, realign, or reinitialize NSG command was attempted before vehicle northings and eastings were entered.

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TABLE IV. TSCD operator error messages – Continued.

Display	Cause
NSG NOT RDY	A polar data request or a realign NSG command was attempted before the NSG was in the Directional-Gyro mode.
TOO MANY	Number entered had too many digits.
TRIG SHORT	Trigger switch in hand control is activated (depressed or electrically shorted) during power-up.
? V EASTINGS	No vehicle eastings value entered.
? V NORTHING	No vehicle northing value entered.
? V ALTITUDE	No vehicle altitude value entered.
? T EASTINGS	No target eastings value entered.
? T NORTHING	No target northing value entered.
? T ALTITUDE	No target altitude value entered.
NO VALD TE/N	No valid target eastings/northing data.
NO VALD VE/N	No valid vehicle eastings/northing data.

3.5.21.4 Lamp test. The LAMP TEST/BRT (lamp test/brightness) key shall provide a menu to examine and change the lamp brightness value, and to perform a lamp test. The power-up value for lamp brightness shall be 9 (full brightness). By entering in numbers between 0 (minimum brightness) and 9, the brightness shall be varied. The lamp test shall test the alphanumeric display by lighting each dot in each character, in groups of four characters, at a rate of one second per group. After testing the display, all status lamps and the trigger lamp on the hand control shall illuminate for three seconds (see 4.6.4.20.4).

3.5.21.5 G/VLLD reticle brightness (GLLD RET BRT) key. The GLLD RET BRT key shall provide a menu to enter and examine the G/VLLD reticle brightness value. A brightness of zero shall set the reticle brightness to the same level as a minimum value of one, but shall cause the numeric display in the G/VLLD optics to be turned off except when the laser is fired. Maximum brightness shall be nine and the power-up brightness value shall be one (see 4.6.4.20.5).

3.5.22 Digital message device (DMD). The FISTV vehicle shall have provisions for accepting either the single channel AN/PSG-2A DMD or the four channel AN/PSG-5 FIST DMD, along with the DMD mount (see 4.6.4.21).

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3.5.22.1 DMD display. The DMD shall be able to display a mode menu as shown in figure 2, when the power is turned on or when the mode key is depressed, while power is on. Alphabetic, numeric and function keys, when pressed, shall display their respective characters when the operator (OPR) mode is chosen. The display bright (DSP BRT) and keyboard bright (KYBD BRT) controls shall rotate smoothly without binding and shall provide continuous control from total darkness to maximum brightness (see 4.6.4.21.1).

3.5.22.2 DMD message transmission. The DMD in the FISTV vehicle shall be capable of transmitting and receiving messages with another DMD at a base or remote station location. The XMIT key shall cause a message to be transmitted. The MSG key shall allow received messages to be displayed. The MSGl and MSG2 indicators shall illuminate respectively when the DMD has received a message. The FIST DMD shall be capable of receiving a designate command from a remote DMD or other equivalent device. This command shall illuminate a green lamp in the G/VLLD eyepiece display (see 4.6.4.21.2).

3.5.22.3 DMD/TSCD interface. The TSCD shall interface with the DMD to permit automatic transmission of a polar data word from the TSCD to the DMD whenever the G/VLLD is fired in a ranging mode (see 4.6.4.21.3).

3.5.23 Ground/vehicular laser locator and designator. The G/VLLD shall be capable of being fired by the trigger switch on the targeting station hand control. The G/VLLD shall not fire if a GLLD Code has not been entered into the TSCD or if a fire interrupt has been detected. When G/VLLD power is on and the dust cover is not opened, the audio alarm shall be activated (see 4.6.4.22).

3.5.23.1 G/VLLD remote control. A remotely operated dust cover shall be provided for the G/VLLD objective lens. The cover shall be capable of being placed over the lens or removed from the FOV by means of a manually actuated remote control system accessible to the targeting station operator (see 4.6.4.22.1).

3.5.24 North-seeking gyrocompass (NSG).

3.5.24.1 NSG lamps. The NSG ALIGN lamp on the TSCD front panel shall flash, upon power up, until vehicle coordinates are entered into the TSCD. The NSG ALIGN lamp shall be illuminated for no more than 10 minutes and 20 seconds when initializing and no more than five minutes and 20 seconds when re-aligning. Once the NSG is ready to use, the NSG READY lamp shall illuminate (see 4.6.4.23.1).

3.5.25 Electrical system. The FISTV vehicle electrical system shall include all components required for generation, conversion, distribution, and control of electrical power. The electrical system shall be in accordance with the vehicle electrical power characteristics as defined by MIL-STD-1275. The equipment bus shall provide nominal power between 20 and 30 V dc. Communication and electrical circuits shall maintain continuity from end to end, without any shorts from one pin to the other, or from each pin to the connector shell, to assure proper function of communication and other electrical equipment (see 4.6.4.24).

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3.5.25.1 Winterization equipment. The FISTV vehicle shall make provisions for installation of an engine coolant and battery heater kit. This kit shall enable the vehicle to start after a cold soak period of not less than eight hours nor more than 12 hours at -65°F (-54°C) (see 4.6.4.24.1).

3.5.25.2 Emergency power. A back-up emergency power provision shall be available which allows the equipment and vehicle batteries to be connected in parallel. This shall enable starting the engine and operating the hydraulic pump with insufficiently charged vehicle batteries and allow use of the electrical and electronic equipment from vehicle batteries when equipment batteries are insufficiently charged. The emergency power shall be controlled by the master switch unit at the drivers station (see 4.6.4.24.2).

3.5.25.3 Lights. The lights, as specified on applicable drawings, shall operate throughout all vehicle operating conditions (see 4.6.4.24.3).

3.5.25.4 Slip ring. The FISTV vehicle slip ring shall provide uninterrupted continuity of communications, power and signal circuits throughout continuous bi-directional rotation of 360 degrees. The operation of the slip ring shall not introduce noise or static into the communications system, which would impair intercommunication intelligibility (see 4.6.4.24.4).

3.5.26 Communications. The communications station shall make provisions for installation of one AN/VRC-46 radio set, three AN/GRC-160 radio sets, one AN/PSG-2A or AN/PSG-5 DMD, one COMSEC/TSEC KY-57 device, one Position Location Reporting System (PLRS) and one intercommunication distribution unit. Each of the four crew operating stations shall be provided with an intercommunication control unit. The communication system shall provide intelligible conversation during 360 degrees azimuth rotation of the turret and during the cycling of the hydraulic pump (see 4.6.4.25).

3.5.26.1 Intercommunication set. The intercommunication distribution and control units shall permit each of the four operating stations: driver, targeting station, observation and communications, to selectively monitor the intercommunication system and any combination of the four radio receivers while selectively transmitting on any of the four radio transmitters as well as the intercommunications system. The intercommunication control unit for the communication station (commander) shall be connected so that by means of a switch on the intercom distribution unit it may be the only such station with voice transmission via the selected radio. This switch shall allow for commander and crew (CDR + CREW), commander only (CDR ONLY), or LISTEN ONLY modes of operation. It shall be possible to transmit simultaneously on all four radios when they are assigned to different frequencies. The noise level of the targeting station intercom shall not unreasonably degrade communications. Each crew station shall have volume control that adjusts audio level (see 4.6.4.25).

3.5.27 Hydraulic system.

3.5.27.1 Hydraulic fluid leakage. Leakage at the hydraulic lines, connections and reservoirs shall not exceed a droplets condition as defined in 6.4.1 (see 4.6.4.26.1).

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3.5.27.2 Hydraulic fluid pressure. The azimuth and elevation drive systems shall be supplied with regulated hydraulic pressure within the range of 1400 to 1500 pounds per square inch (psi) [9.65 to 10.3 millipascals (MPa)]. The accumulator nitrogen per charge shall be 1000 ± 100 psi (6.9 ± 0.7 mPa) at 72°F (22.2°C) with zero hydraulic pressure on the system. A manual hydraulic pump shall be provided for the manual backup mode, as specified in 3.5.27.3.1. Hydraulic pressure loss shall not exceed 300 psi (2.07 MPa) in seven minutes for a non-active, erect targeting head and fully pressurized system (see 4.6.4.26.1).

3.5.27.3 Hydraulic pump. When the accumulator pressure is less than 1400 psi (9.65 MPa), the hydraulic pump shall turn on. In not more than 12 seconds, the accumulator pressure shall reach 1800 ± 100 psi (12.4 ± 0.7 MPa) and the pump shall turn off (see 4.6.4.26.2).

3.5.27.3.1 Manual hydraulic pump. A manual hydraulic pump shall be provided which shall enable an operator to achieve hydraulic system pressure of 1750 psi (12.07 MPa). The manual hydraulic pump shall enable the operator to raise the accumulator pressure from 100 psi (0.7 MPa) to 1500 psi (10.3 MPa) in 150 pump cycles or less. The required force per pump cycle shall not exceed 60 lbs (267 N). Use of the manual hydraulic pump shall allow the targeting head to be slewed in azimuth and elevation when there is insufficient power available to electrically increase hydraulic pressure (see 4.6.4.26.2.1).

3.5.28 Controls and adjustment mechanisms. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all operating conditions. The adjustment mechanisms, including observer's, driver's, targeting station and radio operator's seats, shall function properly and maintain adjustment settings during all vehicle operating conditions (see 4.6.4.27).

3.5.28.1 Hand control. The targeting station hand control shall provide the operator with control of the targeting station azimuth and elevation rates of motion, laser firing and an optical sight selection capability. It shall provide a red indicator light that illuminates when the trigger circuit is closed. The minimum control handle rotation from the neutral position to produce turret motion in either right or left traverse, or in elevation or depression, shall not exceed 4 degrees. The control handle shall return to the neutral position when released at any position and shall not produce any drift in azimuth or elevation (see 4.6.4.27.1).

3.5.29 Safety interrupt and override. Safety interlocks shall be provided to protect the crew and equipment when operating the targeting head and firing the laser. An override capability shall be provided which will permit target tracking and laser firing in the event of operational necessity. When this override is used, an audible alarm shall sound. The interrupt conditions and resulting functions which shall be allowed are summarized in table V (see 4.6.4.28).

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TABLE V. Targeting station safety interrupt condition.

Condition	Function enabled		
	Laser fire	Stow	Turret motion 1/
Driver's Hatch Open	No #	Yes	No # 2/
Targeting Station Hatch Open	No #	Yes	No #
Cargo Hatch Open Partially	No #	No	No #
Cargo Hatch Full Open	No #	Yes	Yes
Stow Position	No	N/A	No
G/VLLD Mount Unlocked	Yes	Yes	No #
Erect Locks Not Engaged	Yes	Yes	No #
No Elevation Stow	Yes	No	Yes
No Azimuth Stow	Yes	No	Yes
WFOV Selected	No	Yes	Yes
Trigger Switch Shorted	No	Yes	Yes

1/ Turret motion consists of track, slew and elevation.
2/ # Indicates override will enable function.

3.5.30 Operating conditions.

3.5.30.1 Climatic conditions. The FISTV vehicle shall operate in ambient air temperatures of 125°F (51.7°C) to minus 25°F (-31.7°C) (see 4.6.4.29.1).

3.5.30.2 Shock and vibration. The FISTV vehicle shall not be damaged, nor shall its performance be impaired, after exposure to shock and vibration experienced during the operating conditions specified herein (see 4.6.4.29.1).

3.5.31 Survivability.

3.5.31.1 Armament. The armament provisions of the FISTV vehicle shall consist of a traversing rail and pintle gun mount for the 7.62 mm M60 machine gun. The rail shall be mounted on the targeting station cupola. A locking device shall be provided to secure the mount at any position along the traversing rail. In the travel mode, the machine gun shall (see 4.6.4.30.1):

a. Not interfere with any position of the targeting head or targeting station.
b. Permit continuous targeting station rotation without interference with the M981 chassis or panoramic telescope.
c. Remain stowed under the conditions defined in the mission profile (see 3.5.3).

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3.5.31.1.1 Pintle mount. The breakaway force, at the pintle mount, needed to move the M60 machine gun from a locked position shall not be less than 45 lbs (200 N). The force required after breakaway to move the mount along the traversing rail without binding shall be not more than 15 lbs (67 N) (see 4.6.4.31.1.1).

3.5.31.2 Ventilated face mask (VFM) system. The M1Al-19 ventilated face mask collective filter system shall be provided to allow an M25-A1 ventilated face mask to be hooked up at each of the four crew stations. The air flow rate at each VFM air hose shall be 3.0-4.5 cubic feet per minute (cfm) [14.1-21.2 cubic meter per second (cu m/sec)] at 2.0-4.8 inches water gage static pressure (0.5-1.2 kPa). The heater control shall warm the air to above the vehicle ambient temperature after one minute of operation (see 4.6.4.30.2).

3.5.31.3 Smoke grenade system. The FISTV vehicle shall accept a self-screening protective smoke system consisting of two grenade launchers, each holding four smoke grenades. When the smoke grenade launcher power control selector is on, the indicator illuminated, and the firing switches activated, supply voltage at the launcher grenade terminals shall correspond to vehicle battery voltage (see 4.6.4.30.3).

3.5.31.4 Electromagnetic radiation (EMI/EMC). The FISTV vehicle shall be radio interference suppressed in accordance with tactical vehicle provisions for class IIIA of MIL-STD-461, notice 4, except as noted in b and c below. GFE equipment not meeting this requirement shall not be cause for noncompliance (see 4.6.4.30.4).

Specific areas of required compliance are:

a. For modified M113A2 chassis:
   RE 05 [14 kilohertz (kHz) to 16 megahertz (MHz)]
   CE 07 (1.5 MHz to 65 MHz)
   For M901 (ITV) common items:
   RE 02 (14 kHz to 1000 MHz)
   CE 04 (50 kHz to 50 MHz)
   For M981 (FISTV) peculiar items:
   CE 01 Conducted Emission, Power Leads (30 Hz-50 kHz)
   CE 04 Conducted Emission, Power Leads (50 kHz-50 MHz, Boadband noise only).
   CE 07 Conducted Emissions, Power Leads, Spikes, Time Domain (1.5 MHz-65 MHz).
   CS 01 Conducted Susceptibility, Power Leads (30 kHz-50 kHz)
   CS 02 Conducted Susceptibility, Power Leads (50 kHz-400 MHz)
   CS 06 Spike, Power Leads (50 kHz-400 MHz)
   RE 05 Broadband Radiated Emissions (14 kHz-16 MHz)
   RS 02 Magnetic Induction Field-Spike
   RS 03 Radiated Susceptibility (10 kHz-12.4 GHz)
b. Mission essential equipment shall be tested in accordance with the applicable test of MIL-STD-461, notice 4 for that type of equipment.
c. The vehicle horn and turret hydraulic pump motor are excluded.

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3.5.31.5 Electromagnetic pulse (EMP). All FISTV vehicle shall not be damaged by exposure to the EMP conditions defined in the M981 Nuclear Survivability Criteria (see 4.6.4.30.5).

3.6 Painting, finish, marking and data plates.

3.6.1 Painting. Unless otherwise specified (see 6.2), the exterior surface of all interior vehicle components; except cables, optics, displays, radios, and vision blocks, shall be painted light green, No. 24533 in accordance with FED-STD-595, epoxy-polyamide chemical agent resistant coating in accordance with MIL-C-22750, type I, except the intercom units shall be painted olive drab, No. 24087. The exterior of all exterior components; excluding optics, antennas and vision blocks, shall be painted with chemical agent resistant aliphatic polyurethane coating in accordance with MIL-C-46168, Color No. 383 green. All aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541 and primed in accordance with MIL-P-23377, type I. All steel interior surfaces shall be primed in accordance with MIL-P-52192 (see 4.7).

3.6.2 Electro-mechanical finish. Unless otherwise specified, surface treatments, coatings and finishes for the FISTV vehicle peculiar electro-mechanical equipment and devices shall conform to MIL-P-11268 (see 4.7).

3.6.3 Marking. The FISTV vehicle exterior markings shall be lusterless black, Color No. 37038 of FED-STD-595, aliphatic polyurethane in accordance with MIL-C-46168, except for safety markings, and shall conform to MIL-STD-642. Interior markings shall be Black No. 17038 in accordance with MIL-C-22750. Parts and components shall be marked in accordance with MIL-STD-130 (see 4.7).

3.6.3.1 Safety markings. Unless otherwise specified (see 6.2), safety hazard plates and decals for the FISTV vehicle and component assemblies shall be posted in accordance with applicable drawings (see 4.7).

3.6.4 Data plates. The data plates shall be in accordance with the requirements of MIL-P-514, or applicable drawings (see 4.7).

3.7 Workmanship. Workmanship shall be of a quality to assure that the vehicle and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use. In addition to general appearance, the defects listed in table VIII shall be considered to be cause for failure. Electric wires and hydraulic lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. No workmanship condition shall exist which may create a safety hazard to operating or maintenance personnel. Particular attention shall be paid to marking of parts, plating, machine-screw assemblage, welding, brazing and freedom of parts from burrs and sharp edges (see 4.8).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designated inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662. Where compliance with MIL-STD-45662 is not possible, special calibration requirements shall be prepared for approval by the acquisition activity.

4.2 Classification of inspection.

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).
   3. Production sample inspection (see 4.4.3).

b. Quality conformance inspection.
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).
   4. Comparison tests (see 4.5.3.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a.	Temperature:	73 $\pm$ 18 °F (22.8 $\pm$ 10 °C)
b.	Barometric pressure:	28.5 (+2.0, -3.0) inches mercury (Hg) (94.6 $\pm$ 8.4 kPa)
c.	Relative humidity:	50 $\pm$ 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction, initial production, or production vehicles samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply vehicles that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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4.4.1 Preproduction inspection. The first two preproduction vehicles, representative of units to be manufactured in quantity and constructed with tools and equipment normally used in quantity production, shall undergo inspection at a location approved by the Government prior to full scale production. This inspection shall determine conformance to the requirements of this specification. Inspection shall consist of examinations specified in 4.5.2 and tests as specified in table VII.

4.4.2 Initial production inspection. When specified (see 6.2) an initial production inspection shall be performed on no less than two vehicles, selected at random after the contractor considers the vehicles produced are representative of production parameters. Inspection shall consist of examination as specified in 4.5.2 and all tests as specified in table VII. The vehicle shall be road tested for 2250 miles (3621 km) as specified in table VI.

TABLE VI. 2250 mile test.

Course	Mileage (percentage)
Hard Surface	900 miles (1448 km) (40%)
Dirt/Gravel	450 miles (724 km) (20%)
Level Cross Country	450 miles (724 km) (20%)
Hilly Cross Country	450 miles (724 km) (20%)

4.4.3 Production sample inspection. A vehicle produced from production tooling and assembled using production methods shall be selected at random from the first month production or the first twenty built, and subjected to the production sample inspection.

4.4.3.1 In-process examination. During fabrication of the initial production vehicle, an in-process examination be conducted by representatives of TACOM product assurance directorate to evaluate conformance of materials and workmanship to specified requirements. Examination will be made at the contractor or subcontractor facility prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records will be evaluated during this examination.

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TABLE VII. Classification and location of inspections.

Title	Requirement	Test	First article			Quality conformance		
			Preprod vehicle GPG	Initial prod GPG	Prod sample contr	Accept contr	Control contr	Cmprsn GPG
Material, design and construction	3.2 thru 3.3.6.2	4.6.1 thru 4.6.2	X	X	X		X	
Weight	3.4.1	4.6.3.1				X		
BII	3.4.2	4.6.3.2	X	X	X			X
Component removal	3.4.3	4.6.3.3	X		X	X		X
Transportability	3.4.4	4.6.3.4	X	X				X
Welding repairs	3.4.5	4.6.3.5	X	X		X		X
Performance	3.5	4.6.4	X	X	X	X	X	X
Reliability	3.5.1.1	4.6.4.1	X	X				X
Maintainability	3.5.1.2	4.6.4.1	X	X				X
MAXTTR	3.5.1.2.1	4.6.4.1	X	X				
MR	3.5.1.2.2	4.6.4.1	X	X				
Automotive operation	3.5.2	4.6.4.1	X	X		X		X
Mission profile	3.5.3	4.6.4.2.1	X	X		X		X
Speed	3.5.4	4.6.4.3	X	X		X		X
Braking	3.5.5	4.6.4.4	X	X	X	X	X	X
Holding	3.5.6	4.6.4.5	X	X	X	X	X	X
Shallow water crossing	3.5.7	4.6.4.6	X	X	X	X		X
Amphibious operation	3.5.8	4.6.4.7	X	X	X	X	X	X
Leakage	3.5.9.1	4.6.4.8.1	X	X		X		X
Bearing seals	3.5.9.2	4.6.4.8.2	X	X		X		X
Trench crossing	3.5.10	4.6.4.9	X	X				X
Vertical obstacles	3.5.11	4.6.4.10	X	X				X
Climbing	3.5.12	4.6.4.11	X	X			X	X
Acceleration	3.5.13	4.6.4.12	X	X		X		X
Hatches	3.5.14	4.6.4.13	X	X		X		X
Hatch transition time	3.5.14.1	4.6.4.13.1	X	X		X		X
Ramp system	3.5.15	4.6.4.14	X	X		X		X
Bilge pumps	3.5.16	4.6.4.15	X	X		X	X	X
Trim vane	3.5.17	4.6.4.16	X	X		X		X
Turning radius	3.5.18	4.6.4.17	X	X		X	X	X
Vision devices	3.5.19.1	4.6.4.18.1	X	X		X		X

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TABLE VII. Classification and location of inspections.

Title	Requirement	Test	First article			Quality conformance		
			Preprod vehicle GPG	Initial prod GPG	Prod sample contr	Accept contr	Control contr	Comprsn GPG
Panoramic telescope	3.5.19.2	4.6.4.18.2	X	X		X		X
Night vision	3.5.19.3	4.6.4.18.3	X	X		X		X
Night sight controls	3.5.19.3.1	4.6.4.18.3.1	X	X		X	X	X
Image transfer assembly (ITA)	3.5.19.4	4.6.4.18.4	X	X		X		X
Sight selection	3.5.19.4.1	4.6.4.18.4.1	X	X		X		X
Image movement	3.5.19.4.2.1	4.6.4.18.4.2.1	X	X		X		X
Internal atmosphere	3.5.19.4.2.2	4.6.4.18.4.2.2	X	X		X		X
Boresight	3.5.19.5	4.6.4.18.5	X	X		X		X
Targeting station	3.5.20	4.6.4.19	X	X		X		X
Azimuth slew mode	3.5.20.1	4.6.4.19.1	X	X		X		X
Azimuth track mode	3.5.20.1.1	4.6.4.19.1.1	X	X		X	X	X
Elevation (maximum rate)	3.5.20.2	4.6.4.19.2	X	X		X	X	X
Elevation (minimum rate)	3.5.20.2.1	4.6.4.19.2.1	X	X		X	X	X
Azimuth and elevation positional capability	3.5.20.2.2	4.6.4.19.2.2				X	X	X
Pointing and tracking accuracy	3.5.20.3	4.6.4.19.3	X	X		X	X	X
Target locating accuracy	3.5.20.4	4.6.4.19.3	X	X		X	X	X
Position retention	3.5.20.5	4.6.4.19.4	X	X		X	X	X
Slope operation	3.5.20.6	4.6.4.19.5	X	X		X		X
Erection drive	3.5.20.7	4.6.4.19.6	X	X		X		X
Erection position	3.5.20.7.1	4.6.4.19.6.1	X	X		X		X
Erection lock cylinders	3.5.20.7.2	4.6.4.19.6.2	X	X		X		X
Manual erect	3.5.20.7.3	4.6.4.19.6.3	X	X		X		X
Erection motor speed and time	3.5.20.7.4	4.6.4.19.6.4	X	X		X		X

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TABLE VII. Classification and location of inspections.

Title	Requirement	Test	First article			Quality conformance		
			Preprod vehicle GPG	Initial prod GPG	Prod sample contr	Accept contr	Control contr	Cmprsn GPG
Power	3.5.20.7.5	4.6.4.19.6.5	X	X				X
Stow mode	3.5.20.7.6	4.6.4.19.6.6	X	X		X		X
TSCD display	3.5.21.1	4.6.4.20.1	X	X		X		X
TSCD indicator	3.5.21.2	4.6.4.20.2	X	X		X		X
System test	3.5.21.3	4.6.4.20.3	X	X		X		X
Lamp test	3.5.21.4	4.6.4.20.4	X	X		X		X
G/VLLD reticle brightness key	3.5.21.5	4.6.4.20.5	X	X		X		X
DMD	3.5.22	4.6.4.21	X	X		X		X
DMD display	3.5.22.1	4.6.4.21.1	X	X		X		X
DMD message transmission	3.5.22.2	4.6.4.21.2	X	X		X		X
DMD/TSCD interface	3.5.22.3	4.6.4.21.3	X	X		X		X
G/VLLD	3.5.23	4.6.4.22	X	X		X		X
G/VLLD remote controls	3.5.23.1	4.6.4.22.1	X	X		X		X
NSG lamps	3.5.24.1	4.6.4.23.1	X	X		X		X
Electrical system	3.5.25	4.6.4.24	X	X		X		X
Winterization equipment	3.5.25.1	4.6.4.24.1.1	X	X				
Emergency power	3.5.25.2	4.6.4.24.2	X	X		X		X
Lights	3.5.25.3	4.6.4.24.3	X	X		X		X
Slip ring	3.5.25.4	4.6.4.24.4	X	X		X		X
Communication	3.5.26	4.6.4.25	X	X	X	X		X
Intercommunications set	3.5.26.1	4.6.4.25	X	X	X	X		X
Hydraulic fluid leakage	3.5.27.1	4.6.4.26.1	X	X		X	X	X
Hydraulic fluid pressure	3.5.27.2	4.6.4.26.2	X	X	X	X		X
Hydraulic pump	3.5.27.3	4.6.4.26.3	X	X	X	X		X
Manual hydraulic pump	3.5.27.3.1	4.6.4.26.3.1	X	X	X	X		X
Controls and adjustment mechanisms	3.5.28	4.6.4.27	X	X	X	X		X

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TABLE VII. Classification and location of inspections.

Title	Requirement	Test	First article			Quality conformance		
			Preprod vehicle GPG	Initial prod GPG	Prod sample contr	Accept contr	Control contr	Cmprsn GPG
Hand control	3.5.28.1	4.6.4.27.1	X	X		X		X
Safety interrupt and override	3.5.29	4.6.4.28	X	X	X	X		X
Climatic conditions	3.5.30.1	4.6.4.29.1	X	X				X
Shock and vibration	3.5.30.2	4.6.4.29.1	X	X				X
Armament	3.5.31.1	4.6.4.30.1	X	X		X		X
Pintle mount	3.5.31.1.1	4.6.4.30.1.1	X	X		X		X
VFM	3.5.31.2	4.6.4.30.2	X	X		X		X
Smoke grenade system	3.5.31.3	4.6.4.30.3	X	X		X		X
EMI/EMC	3.5.31.4	4.6.4.30.4	X	X				X
EMP	3.5.31.5	4.6.4.30.5	X	X				X
Painting, finish, marking and data plates	3.6.1 thru 3.6.4	4.7	X	X				X
Workmanship	3.7	4.8	X	X		X		X
Preservation, packaging, and vehicle processing	Section 5	4.9	X	X				X

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4.4.3.2 Completed production sample contractor inspection. The production sample vehicle shall be road tested and inspected by the contractor as specified in 4.5.3.1, to determine conformance to contract and specifications. After inspection, the contractor shall submit the vehicle (and all inspection records and certifications) to the responsible Government representative at contractor plant for preliminary examination.

4.4.3.3 Preliminary examination. The responsible Government representative shall conduct a preliminary examination as specified in 4.5.2, of the production sample vehicle.

4.4.3.4 Provisional inspection. Provisional inspection of the production sample vehicle shall be conducted jointly by representatives of TACOM product assurance directorate and the responsible Government representative. The contractor shall provide any required assistance. Testing will be as specified in 4.5.3.1. Contractor shall make available his inspection plan, records, and certifications pertinent to the vehicle and components.

4.4.3.5 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor. Failure of the contractor to correct defects promptly shall be cause for suspension of acceptance of vehicles until corrective action has been approved by the Government.

4.4.3.6 Vehicle disposition. On completion of the sample production vehicle inspection, the vehicle shall remain at the manufacturing facility as a production sample and shall be released at the discretion of the Government. The contractor shall service and maintain the vehicle during this period.

4.4.3.7 Final approval and acceptance. Final approval and acceptance of the initial production vehicle(s) shall be withheld until production sample vehicle is accepted (see 4.3).

4.4.4 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prime facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all vehicles from an identifiable production period submitted at one time by the contractor to determine conformance with acceptability criteria.

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4.5.1.2 Sampling for examination. Before sampling commences by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.3) to allow normal sampling in accordance with MIL-STD-105. When this has been satisfactorily accomplished, representative sample(s) shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105 for the purpose of visual, and dimensional and primary functional examinations.

4.5.2 Examinations. Vehicle examinations shall consist of visual, dimensional, and primary functional characteristics to determine conformance to the applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the acceptable quality levels (AQL'S) shown in table VIII.

4.5.2.1 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring in five consecutive lots or more within a 30 day period, shall be added to the minor classification of defects with no increase in AQL.

4.5.2.2 Failure of examination. If a vehicle fails to pass any examination specified herein, the Government shall stop acceptance until evidence is provided by the contractor that corrective action has been taken.

4.5.2.3 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on deficiency sheet. Defects noted on a deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor representative to classify the defects in accordance with the classification of defects (see table VIII) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects.

4.5.2.3.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

4.5.2.3.2 Recurring minor defects. A minor defect is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

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TABLE VIII. Classification of defects.

Category	Defect	Method of examination
Major	AQL 25 Defects/100 units	
101	Stowage, damage clearance improper (see 3.4.2).	Visual and functional
102	Welding malfunction, damage, structural defects, weld repairs without authorization or beyond scope of authorization (see 3.4.5).	Visual and functional
103	Vehicle performance, improper mounting, leaks 1/ malfunction, damage, improper clearance (see 3.5.2).	Visual and functional
104	Sealing, malfunction, damage leaks (see 3.5.9).	Visual and functional
105	Bearings, malfunction, damage improper fit (see 3.5.9.2).	Visual and functional
106	Hatches, malfunction, nonconform, leaks damage, (see 3.5.14).	Visual and functional
107	Ramp system, malfunction, leaks, damage, improper fit (see 3.5.15).	Visual and functional
108	Bilge pumps, malfunction, damage (see 3.5.16).	Visual and functional
109	Trim vane, malfunction (see 3.5.17).	Visual and functional
110	Vision devices, malfunction, damage (see 3.5.19).	Visual and functional
111	Targeting station, erection arm, cupola; malfunction, damage, improper clearance, structural and welding defects (see 3.5.20).	Visual and functional
112	AZ and EL drive system, malfunction, damage (see 3.5.20.2.2).	Visual and functional
113	TSCD, malfunction, damage, improper indicators (see 3.5.21).	Visual and functional
114	Electrical system, components, circuits and communication system malfunction, improper improper lighting sequence (see 3.5.25). 2/	Visual and functional
115	Hydraulic control system, malfunction, damage leaks (see 3.5.27).	Visual and functional
116	Manual hydraulic pump, malfunction, damage, improper clearance (see 3.5.27.3.1)	Visual and functional
117	Controls malfunction, improper clearance (see 3.5.28).	Visual and functional
118	Safety interlocks, malfunction, damage (see 3.5.29).	Visual and functional

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TABLE VIII. Classification of defects – Continued.

Category	Defect	Method of examination
Minor	AQL 150 Defects/100 units	
201	Stowage, improper assembly or installation (see 3.4.2).	Visual and functional
202	Weldment, improper assembly or installation (see 3.4.5).	Visual and functional
203	Vehicle components, improper assembly or installation (see 3.5.2).	Visual and functional
204	Sealing, improper assembly or installation (see 3.5.9).	Visual and functional
205	Bearings, improper assembly or installation (see 3.5.9.2).	Visual and functional
206	Hatches, improper assembly, installation, dessicant (see 3.6.15).	Visual and functional
207	Ramp system, improper assembly or installation (see 3.5.15).	Visual and functional
208	Bilge pump, improper assembly or installation (see 3.5.16).	Visual and functional
209	Trim vane, improper assembly or installation (see 3.5.17).	Visual and functional
210	Vision devices, improper assembly, installation dessicant (see 3.5.l9).	Visual and functional
211	Targeting station, improper fits, adjustments, installation, welds, seals pressure switch (see 3.5.20).	Visual and functional
212	AZ and EL drive system, improper assembly or installation (see 3.5.20.2.2).	Visual and functional
213	TSCD, improper assembly or installation (see 3.5.21).	Visual and functional
214	Electrical system, improper assembly or installation (see 3.5.25).	Visual and functional
215	Hydraulic control system, improper assembly, installation, fluid level, pressure (see 3.5.27).	Visual and functional
216	Controls, improper assembly or installation (see 3.5.28)	Visual and SIE 3/
217	Safety interlocks, improper assembly or installation (see 3.5.29).	Visual and functional
218	Markings and data plates, improper location, size, missing, defective (see 3.6.3 thru 3.6.4).	Visual
219	Workmanship, improper installation, fit, adjustment, welds, scale adhesion, hardware or sheet metal (see 3.7).	Visual and functional

1/ Any fuel leakage constitutes a special defect and shall result in total inspection of the lot for this defect.
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE = Standard Inspection Equipment.

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4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests. Each vehicle shall be operated a distance of one mile or more by the contractor prior to performance of acceptance tests and shall be inspected in 4.5.2. Acceptance tests shall include all acceptance tests specified in table VII. Except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled and serviced with fuel, lubricants, and fluids as specified. Vehicles need not be loaded with rated payload, except where specified. After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage or other deficiencies.

4.5.3.2 Control tests. Control tests shall be performed on the first ten vehicles produced. Ten percent of the next 90 vehicles and one percent of all production vehicles after 100 shall undergo control testing. At least one vehicle per month shall be selected for test.

4.5.3.2.1 Failure. Failure of any vehicle to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5.3.3 Comparison test. The Government may select a vehicle at any time during the contract production period and subject it to Government proving ground testing. A comparison test shall consist of all acceptance tests and those tests identified as comparison tests.

4.5.3.3.1 Failure. Comparison test vehicle(s) deficiencies detected during, or as a result of the comparison test, may be cause for rejection of subsequent produced vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the comparison test shall be prima facie evidence that all vehicles currently or subsequently produced are similarly deficient, unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.3.6.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance shall be determined by examination of the characteristics for defects listed in table VIII. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

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4.6.3 Physical characteristics.

4.6.3.1 Weight. The FISTV vehicle shall be weighed, combat loaded and the weight shall conform to 3.4.1.

4.6.3.2 Basic issue items (BII). To determine conformance to 3.4.2, all exterior and interior BII, GFE, or its simulated equivalent shall be stowed (combat loaded) on the selected vehicle, in the spaces provided. It shall be verified that they do not interfere with the operation of the vehicle and its components in anyway.

4.6.3.3 Component removal. To determine conformance to 3.4.3, a four man crew shall be used to dismount and transport to 50 ± 5 m (165 ± 15 ft) from the vehicle all components specified for dismounted operation (see table I). At the completion of this test, the same equipment shall be transported and remounted. The dismount and remount times shall be measured and shall not exceed the specified requirements.

4.6.3.4 Transportability. To determine conformance to 3.4.4, the specified components shall be removed and the targeting head shall be manually cranked down into a low stow transport position. Resultant height of the vehicle shall be measured.

4.6.3.5 Welding repairs. To determine conformance to 3.4.5, all welding repairs and burning operations shall be examined to determine conformance that slag has been removed and approved procedures have been followed.

4.6.4 Performance. To determine conformance to 3.5, except as otherwise specified herein, testing shall be performed on a smooth level, hard surface free of loose material.

4.6.4.1 Reliability and maintainability. Conformance to 3.5.1 through 3.5.2, shall be demonstrated on the vehicle undergoing the 2250 mile (3621 km) test (see table VI) during the initial production test.

4.6.4.2 Mission profile test. A mission profile test shall be performed upon completion of acceptance tests specified in 4.5.3.1. The mission profile test consists of traveling a total of 54 miles (87 km) in accordance with the mission profiles detailed in 4.6.4.2.1.

4.6.4.2.1 Mission profile. To determine conformance to 3.5.3, two mission profiles shall be performed. Prior to each mission profile the pre-operational phase (see 4.6.4.2.1.1) shall be performed. Each mission profile consists of three segments. The sequence of one segment shall be as follows:

Drive 9 miles (14.5 km) around test track
Operational scenario (see 4.6.4.2.1.2)
Ranging scenarios #1 - #3 (see 4.6.4.2.1.3)
Designating scenario #1 (see 4.6.4.2.1.4)
Ranging scenarios #4 - #6
Designating scenario #2
Ranging scenarios #7 - #9
Designating scenario #3

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Ranging scenarios #10 - #12
Designating scenario #4
Ranging scenarios #13 - #15
Designating scenario #5

4.6.4.2.1.1 Pre-operational phase. The pre-operational phase shall be performed before each mission profile. The vehicle shall be checked and serviced. The vehicle shall be positioned over a presurveyed area and the head shall be stowed. The NSG shall be initialized and the head shall then be erected and a reference target located. The targeting head shall then be stowed.

4.6.4.2.1.2 Operational scenario. The operational scenarios shall be performed after the first 9 mile (14.5 km) track run. The FISTV vehicle shall be situated over a pre-surveyed position. Erect the targeting head and observe a referenced target. The target data shall be recorded from the TSCD display. The targeting head shall be stowed. The NSG shall be realigned and the time of the realignment shall be recorded. The targeting head shall then be erected, and the first ranging scenario (see 4.6.4.2.1.3) performed.

4.6.4.2.1.3 Ranging scenario. A total of 45 ranging scenario shall be performed during each mission profile. The FISTV vehicle shall be located at a presurveyed location. The vehicle position, northing, easting, and altitude shall be entered into the TSCD. A known target shall be acquired and the azimuth and elevation of the target shall be recorded. The G/VLLD shall be fired (simulated). During the entire mission profile test, the laser inhibit plug shall be installed and when the trigger switches are depressed a reference range of 9760 on the G/VLLD reticle shall be given. The target's polar and simulated range data shall be sent to the DMD and subsequently transmitted to a base station. The targeting station shall then be rotated 360°, clockwise on even-numbered scenarios and counterclockwise on odd-numbered scenarios, varying the speed and elevation. This shall conclude one ranging scenario.

4.6.4.2.1.4 Designate scenario. The designating scenario shall be performed 15 times during a mission profile. The reticle brightness of the G/VLLD shall be varied. When a voice command to designate is received, the G/VLLD shall simulate designating for 20 seconds. The laser inhibit plug shall be installed at all times. The targeting head shall be erected. This completes one designating scenario.

4.6.4.3 Speed. To determine conformance to 3.5.4, the FISTV vehicle shall be driven at speeds up to the specified maximum and observed for performance requirements. To preclude any damage to the targeting head or the vehicle, during head erect travel, the specified distance and speed shall not be exceded.

4.6.4.4 Braking. To determine conformance to 3.5.5, the FISTV vehicle shall be driven and stopped, as specified, at least three times. Stopping distance of the three stops shall be averaged to determine the final result.

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4.6.4.5 Holding. To determine conformance to 3.5.6, the FISTV vehicle shall be driven up the specified grade, brakes applied and vehicle observed for movement, then with brakes locked for parking, the vehicle shall be observed for movement.

4.6.4.6 Shallow water crossing. To determine conformance to 3.5.7, the FISTV vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall not exceed specified amount after 15 minutes of test.

4.6.4.7 Amphibious operation. To determine conformance to 3.5.8, the FISTV vehicle shall be operated amphibiously and performance observed. The amount of water in the hull shall be measured and shall not exceed specified amount.

4.6.4.8 Sealing.

4.6.4.8.1 Leakage. To determine conformance to 3.5.9.1, all hatches shall be closed and all components properly installed. A spray of water shall be directed at the hatches and other external components from an open hose. The spray shall have a line pressure of not less than 35 psi (0.24 MPa) at a rate of 10 gpm (37.85 L/min) for a period of three minutes. Leakage shall not exceed an average of 10 drops per minute into the vehicle.

4.6.4.8.2 Bearing seals. To determine conformance to 3.5.9.2, after fording or amphibious operations or operation in mud, sand, or snow, there shall be no visual evidence of bearing contamination by foreign matter nor leakage of lubricants from leakproof seals. Seals designed to allow exit of flushing lubricant may evidence leakage.

4.6.4.9 Trench crossing. To determine conformance to 3.5.10, the FISTV vehicle shall be operated over trenches of specified depth and width.

4.6.4.10 Vertical obstacles. To determine conformance to 3.5.11, the FISTV vehicle shall be operated over vertical obstacles of specified height.

4.6.4.11 Climbing. To determine conformance to 3.5.12, the FISTV vehicle shall be tested for its specified operational capabilities on a 60 percent forward and reverse grade and on a 30 percent left and right lateral slope.

4.6.4.12 Acceleration. To determine conformance to 3.5.13, the FISTV vehicle shall be operated as specified to determine acceleration time from 0 to 20 mph (32 k/hr).

4.6.4.13 Hatches. To determine conformance to 3.5.14, all hatches shall be operated and examined for functional requirements.

4.6.4.13.1 Hatch transition time. To determine conformance to 3.5.14.1, the FISTV vehicle shall be stationary with driver's targeting station and cargo hatches full open (ramp shall be up). The hatches shall be closed using three personnel and the transition time shall be measured and shall not exceed 20 seconds. The hatches shall then be returned to full open and the transition time shall be measured and shall not exceed 20 seconds.

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4.6.4.14 Ramp system. To determine conformance to 3.5.15, the ramp hydraulic system shall be operated and examined for evidence of leaks. The ramp shall be operated to fully opened and closed positions and observed for specified requirements.

4.6.4.15 Bilge pumps. To determine conformance to 3.5.16, with sufficient water in the vehicle hull, the bilge pumps shall be operated and observed for functional requirements.

4.6.4.16 Trim vane. To determine conformance to 3.5.17, it shall be verified that the trim vane is properly engaged in the extended, retracted and released positions without binding or excessive looseness.

4.6.4.17 Turning radius. To determine conformance to 3.5.18, the FISTV vehicle shall be operated as specified, turned to the right and to the left a full 360 degrees and performance shall be observed.

4.6.4.18 Optics.

4.6.4.18.1 Vision devices. To determine conformance to 3.5.19.1, all vision devices shall be examined for functional requirements.

4.6.4.18.2 Panoramic telescope. To determine conformance to 3.5.19.2, the panoramic telescope shall be examined for functional requirements.

4.6.4.18.3 Night vision. To determine conformance to 3.5.19.3, the night sight shall be inspected to verify specified driving performance. It shall be verified that an M19 periscope is provided.

4.6.4.18.3.1 Night sight controls. To determine conformance to 3.5.19.3.1, the night sight controls shall be connected, operated and disconnected 10 times with no degradation of performance.

4.6.4.18.4 ITA. To determine conformance to 3.5.19.4, the ITA shall be operated and examined for functional requirements.

4.6.4.18.4.1 Sight selection. To determine conformance to 3.5.19.4.1, ITA sight controls shall be verified by selecting the 3X, 13X and night options, and verifying that the optical transfer occurs within the specified times.

4.6.4.18.4.2 Image quality.

4.6.4.18.4.2.1 Image movement. To determine conformance to 3.5.19.4.2.1, the ITA shall be visually examined during the optical tracking tests to verify that the ITA exhibits no random motion of the image that would impair tracking.

4.6.4.18.4.2.2 Internal atmosphere. To determine conformance to 3.5.19.4.2.2, the ITA shall be visually examined to verify that there is no evidence of internal moisture or image fogging.

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4.6.4.18.5 Boresight. To determine conformance to 3.5.19.5, it shall be verified that the night sight and G/VLLD can be collimated within the specified accuracy. It shall be verified that the optical coupler and the ITA are pupil matched.

4.6.4.19 Targeting station. To determine conformance to 3.5.20, the targeting station shall be operated and functional requirements shall be verified.

4.6.4.19.1 Azimuth slew mode. To determine conformance to 3.5.20.1, the hand control shall be used to demonstrate azimuth rate controls in the slew mode. Maximum slew rate shall be verified by measuring time to travel 180 degrees when the hand control is deflected to the maximum displacement in traverse. This shall be done in both directions.

4.6.4.19.1.1 Azimuth track mode. To determine conformance to 3.5.20.1.1, the hand control shall be used to demonstrate the azimuth rate controls in the track mode. The minimum azimuth traverse rate shall be no greater than 0.1 milliradians per second.

4.6.4.19.2 Elevation (maximum rate). To determine conformance to 3.5.20.2, the hand control shall be used to demonstrate the control of the targeting head elevation rate. Maximum elevation and depression rates shall be checked by measuring the time to travel the full range elevation angle with maximum hand control displacement.

4.6.4.19.2.1 Elevation (minimum rate). To determine conformance to 3.5.20.2.1, the hand control shall be used to demonstrate the minimum elevation rate. The minimum rate shall be 0.1 milliradians per second or less.

4.6.4.19.2.2 Azimuth and elevation positional capability. To determine conformance to 3.5.20.2.2, the hand control shall be used to rotate the targeting station 360 degrees minimum, right and left. To demonstrate maximum elevation and depression, the targeting head shall be operated in elevation in both directions until action is prohibited. Angles shall be measured and recorded.

4.6.4.19.3 Pointing, tracking and target locating accuracy. To determine conformance to 3.5.20.3 and 3.5.20.4, the FISTV vehicle shall be subjected to a tracking test which compares performance of a qualified operator using a tripod mounted G/VLLD to the same operator using a FISTV mounted G/VLLD. Performance shall be as specified. The test shall be repeated at least three times and the scores averaged. The test shall be executed using a target with suitably repeatable movement characteristics in order to assure compatibility of scoring.

4.6.4.19.4 Position retention. To determine conformance to 3.5.20.5, the targeting head shall be erected and a target (presurveyed point) shall be acquired. The hydraulic pump shall be disabled and five minutes shall be allowed to elapse at which time the position (azimuth and elevation) of the head shall be measured and compared with the position at the start of the test. The change shall not exceed the specified value. The NSG may be used as the angle measuring device for this test.

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4.6.4.19.5 Slope operation. To determine conformance to 3.5.20.6, the FISTV vehicle shall be positioned on a 60 percent slope (either forward or reverse). The targeting head shall be erected and shall be rotated 360° clockwise and then 360° counterclockwise. At the conclusion or during the rotations, the targeting station shall be positioned at either ± 90° with respect to the vehicle centerline and the station brakes shall hold against azimuth rotation. This test shall be repeated on a 30 pecent side slope.

4.6.4.19.6 Erection drive. To determine conformance to 3.5.20.7, the targeting station shall be operated to verify the following functional requirements.

4.6.4.19.6.1 Erection position. To determine conformance to 3.5.20.7.1, it shall be visually verified that the targeting head decelerates prior to achieving the erect position, that erection locks are in position, and that the HEAD ERECT and HEAD READY indicators illuminate and that AZ and EL stow indicators are in conformance with table III.

4.6.4.19.6.2 Erection lock cylinders. To determine conformance to 3.5.20.7.2, the targeting head shall be erected and it shall be verified that the erection lock cylinders engage when the hydraulic accumulator indicator reads 1200 psi (8.27 MPa). The hydraulic pressure shall then be bled down and the power shall be removed, and it shall be verified that the lock cylinders remain engaged. The erection lock cylinders shall retract when stow position is selected and the hydraulic pressure is equal to or greater than 1200 psi (8.27 MPa).

4.6.4.19.6.3 Manual erect. To determine conformance to 3.5.20.7.3, the targeting head shall be put in the stow position and manually cranked up to erect position. The torque required shall be less than 20 lbs (89 N). It shall be verified that after accumulator pressure has been manually pumped to 1200 psi (8.27 MPa) or greater and TSCD head position select in ERECT, that erection locks engage when the targeting head erect position is achieved. Verify that insertion of the handcrank disables the erection drive motor.

4.6.4.19.6.4 Erection motor speed and time. To determine conformance to 3.5.20.7.4, the targeting head shall be erected and it shall be verified that the motor shall stop, without hunting or overshooting, when the erection arm reaches the desired position. Erection time shall be measured and shall be eight seconds or less. During hydraulic pump operation, peak current shall be 900 A or less and steady state current shall not exceed 200 A.

4.6.4.19.6.5 Power. To determine conformance to 3.5.20.7.5, it shall be demonstrated that the erection system cannot be operated electrically without vehicle or equipment power. The hand crank shall drive the erection system when insufficient battery power is available.

4.6.4.19.6.6 Stow mode. To determine conformance to 3.5.20.7.6, it shall be verified that all stow indicators are in accordance with table III, and that the targeting head is resting on its stow supports.

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4.6.4.20 TSCD.

4.6.4.20.1 TSCD display. To determine conformance to 3.5.21.1, it shall be verified that up to twelve alphanumeric characters are displayed on the TSCD and that the characters displayed agree with the input characters or the proper error message. See table IV for the allowable error messages.

4.6.4.20.2 TSCD indicators. To determine conformance to 3.5.21.2, the TSCD shall be operated in the vehicle to verify table III.

4.6.4.20.3 System test. To determine conformance to 3.5.21.3, the system test shall be verified by depressing the TEST key on the TSCD and verifying that SYSTEM TEST? is displayed. Then the ENTR key shall be depressed and it shall be verified that TESTING flashes on the display for five seconds and then any error faults shall be displayed as an alphanumeric code.

4.6.4.20.4 Lamp test. To determine conformance to 3.5.21.4, the LAMP TEST/BRT key shall be depressed and it shall be verified that LAMP BRT 9 appears on the display as the power up condition. Numbers between 0 and 9 shall be entered and shall vary the brightness from 0 (minimum) to 9 (maximum). It shall be verified that when the LAMP TEST/BRT key is depressed twice in succession and then entered, all TSCD lamps and displays are tested. Each dot in each character, in groups of four characters, at a rate of one second per group shall illuminate. Then all status lamps and the trigger lamp on the hand control shall illuminate for three seconds.

4.6.4.20.5 G/VLLD reticle brightness key. To determine conformance to 3.5.21.5, it shall be verified that the G/VLLD reticle brightness can be varied by entering values 1 through 9 into the TSCD. To verify the 0 brightness function, the G/VLLD dust cover shall be closed to aid visual inspection, and a simulated G/VLLD firing shall be performed. During all tests involving the G/VLLD, the laser inhibit plug shall be installed, unless otherwise specified.

4.6.4.21 DMD. To determine conformance to 3.5.22, the physical provisions shall be verified.

4.6.4.21.1 DMD display. To determine conformance to 3.5.22.1, the DMD shall be operated to verify the specified requirements.

4.6.4.21.2 DMD message transmission. To determine conformance to 3.5.22.2, the DMD in the FISTV vehicle shall be operated to verify communication operability with a remote DMD or DMD simulator. It shall be verified that a remote DMD or equivalent device can illuminate the green G/VLLD remote designate lamp by sending a designate command.

4.6.4.21.3 DMD/TSCD interface. To determine conformance to 3.5.22.3, the G/VLLD shall be fired in a ranging mode and it shall be verified that the TSCD automatically sends a polar data word to the DMD. This test cannot be performed with the laser inhibit plug installed, so all safety precautions shall be strictly observed.

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4.6.4.22 G/VLLD. To determine conformance to 3.5.23, the G/VLLD shall be operated to verify that it functions according to the specified requirements. Except as otherwise specified herein, any testing involving the G/VLLD requires that the laser inhibit plug be installed. Production G/VLLD simulated firing with inhibit plugs installed shall provide a reference range of 9760.

4.6.4.22.1 G/VLLD remote controls. To determine conformance to 3.5.23.1, the dust cover controls shall be activated and demonstrated to be operational.

4.6.4.23 NSG.

4.6.4.23.1 NSG lamps. To determine conformance to 3.5.24.1, the NSG lamps on the TSCD front panel shall be observed and timed.

4.6.4.24 Electrical system. To determine conformance to 3.5.25, electrical circuits supplying power to vehicle equipment shall be tested to assure continuity and integrity. Circuits tested for continuity prior to vehicle installation, and targeting station power circuitry and circuit functions shall be verified by evidence of functional performance during operational tests. The vehicle radios shall be operated to verify transmission and reception, and it shall be verified by evidence of operational performance that the FISTV electrical system is compatible with the vehicle power supply, which conforms to the requirements of MIL-STD-1275.

4.6.4.24.1 Winterization equipment. To determine conformance to 3.5.25.1, the FISTV vehicle shall sit idle for the length of time specified at -65°F (-54°C) and then it shall be verified that the engine starts after the cold soak period.

4.6.4.24.2 Emergency power. To determine conformance to 3.5.25.2, the following tests shall be done to verify emergency power. With the vehicle battery and emergency power switched ON and the equipment battery OFF, the TSCD shall be powered up and operated. With the equipment battery and emergency power switched ON and the vehicle battery OFF, the engine shall be started.

4.6.4.24.3 Lights. To determine conformance to 3.5.25.3, all FISTV vehicle lighting, including running lights, driver's console lights, azimuth position indicator and driver level illuminator azimuth scale illumination, targeting station indicators, TSCD display, dome lights, utility lights and the gooseneck light, shall be operated during and after road testing through all stages of operation.

4.6.4.24.4 Slip ring. To determine conformance to 3.5.25.4, it shall be verified that the slip ring shall provide continuous contact throughout a bi-directional 360°. It shall also be verified that operation of the slip ring shall not cause enough noise or static to degrade required communications.

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4.6.4.25 Communications. To determine conformance to 3.5.26 and 3.5.26.1, the communications equipment shall be functionally checked. It shall be verified that each intercom station has talk and listen capability over both radio and intercom channels, in accordance with system performance requirements, over 360° of targeting station azimuth rotation. It shall be verified that all four radios can receive and transmit simultaneously on four different frequencies. Communications system noise level shall be monitored by ear during intercom use while the targeting station hydraulic system pump is cycled. The intercom noise level shall not cause excessive reduction of intelligibility of conversation.

4.6.4.26 Hydraulic system.

4.6.4.26.1 Hydraulic fluid leakage and pressure. To determine conformance to 3.5.27.1 and 3.5.27.2, the hydraulic system shall be operated, examined for leakage in excess of a droplet condition (see 6.4), and it shall be verified that the accumulator nitrogen precharge is not less than 900 psi (6.2 MPa) nor greater than 1100 psi (7.58 MPa) at the specified environment. It shall be verified that internal leakage and resultant system pressure loss does not exceed specified limits by verifying that the time interval between hydraulic pump motor activation, when the targeting station is powered up but not active, is not less than seven minutes.

4.6.4.26.2 Hydraulic pump. To determine conformance to 3.5.27.3, it shall be verified, when the accumulator pressure is less than 1400 psi (9.65 MPa) and the power is turned on, that the hydraulic pressurization pump comes on. The pump shall turn off when the accumulator pressure reaches 1800 $\pm$ 100 psi (12.4 $\pm$ 0.7 MPa) and it shall be verified that it takes no more than 12 seconds to reach operating pressure.

4.6.4.26.2.1 Manual hydraulic pump. To determine conformance to 3.5.27.3.1, the targeting station hydraulic system shall be manually pumped to meet the requirements, and it shall be verified that the targeting head can be slewed in azimuth and operated in elevation using only the manual back-up hydraulic pump and equipment power.

4.6.4.27 Controls and adjustment mechanism. To determine conformance to 3.5.28, the devices shall be exercised or otherwise functionally activated and demonstrated to be operational.

4.6.4.27.1 Hand control. To determine conformance to 3.5.28.1, the targeting station hand control shall be deflected to 4 degrees in azimuth and the azimuth angle indicator shall reveal targeting station motion. The test shall be repeated in the opposite direction in azimuth, and similarly for elevation and depression.

4.6.4.28 Safety interrupts and override. To determine conformance to 3.5.29, table V shall be verified. The following tests shall be required for verification.

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a. Stow interlock check. It shall be verified that for the targeting head in the stowed mode the laser cannot be fired and the turret cannot be moved. It shall be verified that the override function does not enable firing of the laser, nor movement of the targeting head.

b. Targeting station interlock and override. It shall be verified that the laser interrupt and weapon hatch indicators come on, the laser cannot be fired and turret motion is inhibited when the targeting station hatch has been opened. The stow position shall be allowed. It shall be verified that the override function enables firing of the laser and turret motion, and sounds the alarm. It shall be verified that turning off the override turns off the audible alarm. It shall be verified that closing the targeting station hatch turns off the laser interrupt indicators.

c. Driver's hatch interlock and override. It shall be verified that the laser interrupt and driver's hatch indicators come on, the laser cannot be fired and turret motion is inhibited when the driver's hatch has been positioned to partial or full open. The stow position shall be allowed. It shall be verified that the override function enables firing of the laser, turret motion and sounds the alarm. It shall also be verified that closing the driver's hatch turns off the audible alarm and laser interrupt indicators.

d. Cargo hatch interlock and override. It shall be verified that the laser interrupt and cargo hatch indicators come on and the laser cannot be fired when the cargo hatch has been opened (partial or full). When the cargo hatch is open (partial), it shall be verified that stow position and turret motion are also inhibited. It shall be verified that the override function enables firing of the laser, turret motion and sounding of the alarm. It shall be verified that turning off the override turns off the audible alarm. It shall be verified that the stow position, with a partial open cargo hatch cannot be overridden. It shall be verified that closing the cargo hatch turns off the laser interrupt indicators.

e. WFOV interlock. It shall be verified that the laser interrupt indicator comes on and the laser cannot be fired when the WFOV has been selected. It shall be verified that the override function does not enable firing of the laser.

f. Shortage trigger switch interlock. It shall be verified that when one or both trigger switches on the hand control unit is depressed or failed during TSCD power up and when there are no fire interrupts and when the G/VLLD power is on, the laser cannot be fired. It shall be verified that a flashing "TRIG SHORT" operator error message shall appear on the TSCD display. It shall be verified that the override function shall not enable firing of the laser.

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g. Stow interlocks. It shall be verified that the targeting station shall not go into stow position when the EL STOW and AZ STOW indicators are not illuminated, indicating that the proper stow position has not been achieved. The override function shall not allow the head to be stowed.

h. G/VLLD mount interlock. It shall be verified that turret motion is not allowed when the G/VLLD mount is unlocked. It shall be verified that the override function shall sound the alarm and allow turret motion.

i. Erect lock interlock. It shall be verified that turret motion is not allowed when the erection locks are not in. It shall be verified that the override function shall sound the alarm and allow turret motion.

4.6.4.29 Operating conditions.

4.6.4.29.1 Climate, shock and vibration. To determine conformance to 3.5.30.1 and 3.5.30.2, the FISTV vehicle shall be operated in specified environments and observed for performance requirements.

4.6.4.30 Survivability.

4.6.4.30.1 Armament. To determine conformance to 3.5.31.1, it shall be verified that the gun pintle mount, when moved across the traversing rail, shall be free of binding. It shall also be verified that the locking device secures the mount at any position along the traversing rail.

4.6.4.30.1.1 Pintle mount. To determine conformance to 3.5.31.1.1, it shall be verified that the mount shall not move with a force of up to 45 lbs (200 N). After breakaway, the force applied parallel to the traversing rail at the machine gun pintle mount shall be measured to be 15 lbs (67 N) or less.

4.6.4.30.2 Ventilated face mask. To determine conformance to 3.5.31.2, the M39 air flow tester shall be used to verify flow rate at each of the four hoses. The heater shall be turned on and the temperature of the air shall be as specified.

4.6.4.30.3 Smoke grenade system test. To determine conformance to 3.5.31.3, the grenade launcher power selector shall be placed “ON” and illumination of the indicator shall be verified. The voltage between grenade terminals on the grenade launcher shall be measured when the firing switches are activated. Voltage shall be as specified.

4.6.4.30.4 EMI/EMC. To determine conformance to 3.5.31.4, the specified tests in MIL-STD-462 shall be performed and compliance shall be verified.

4.6.4.30.5 EMP. To determine conformance to 3.5.31.5, the FISTV vehicle shall be subjected to verification – reference Detailed Test Plan, Nuclear Effects Subtest of Development Test II (PQT-G) of FISTV, Annex A, October 1982 (confidential).

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4.7 Painting, finish, marking and data plates. The FISTV vehicle shall be visually examined to verify conformance to requirements of 3.6.1 through 3.6.4.

4.8 Workmanship. The FISTV vehicle shall be examined to determine conformance to 3.8.

4.9 Preservation, packaging, and vehicle processing inspection. Material and equipment shall be inspected, prior to shipment, to determine conformance to Section 5 of this specification. Sampling shall be as specified in applicable packaging specification reference therein.

5. PACKAGING

5.1 Preparation for delivery. Preparation for delivery and storage shall be in accordance with MIL-T-62327.

6. NOTES

6.1 Intended use. The FISTV (M981) vehicle is intended for, but not limited to fire support operations. The M981 must be air transportable and capable of crossing inland waterways under its own power.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. M981 (FISTV) Drawing 8750031 required.
c. If first article samples are not required (see 3.1).
d. If welding repairs are authorized, and extent allowed (see 3.4.5).
e. If winterization equipment is required (see 3.5.25.1).
f. Painting, protective finish, marking, and data plates, if other than specified (see 3.6).
g. If responsibility for inspection shall be other than as specified (see 4.1).
h. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
i. If Inspection conditions shall be other than as specified (see 4.3).
j. If preproduction inspection is required (see 4.3).
k. If initial production inspection is not required (see 4.4.2).
l. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

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6.4 Definitions.

6.4.1 Leakage. The following definitions shall be used with the classification of defects:

a.	Weep:	Any evidence of fluid beyond the seal.
b.	Seep:	Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c.	Droplet:	Any evidence of fluid beyond the seal that results in the formation of a fluid mass which does not fall.
d.	Drip:	Any evidence of fluid beyond the seal where droplet forms and falls.
e.	Leak:	A steady series of drips.
f.	Excessive leaks:	A steady stream of fluid.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

6.6 Safety precautions.

6.6.1 Radioactive materials. All test personnel shall be cautioned as to the potential hazards of radioactive materials and instructed how to properly dispose of broken night sight lenses (see 3.3.2).

6.7 Supersession data. This military specification supersedes Emerson Electric Company product fabrication specification M981PD-V-62475, dated 28 August 1984.

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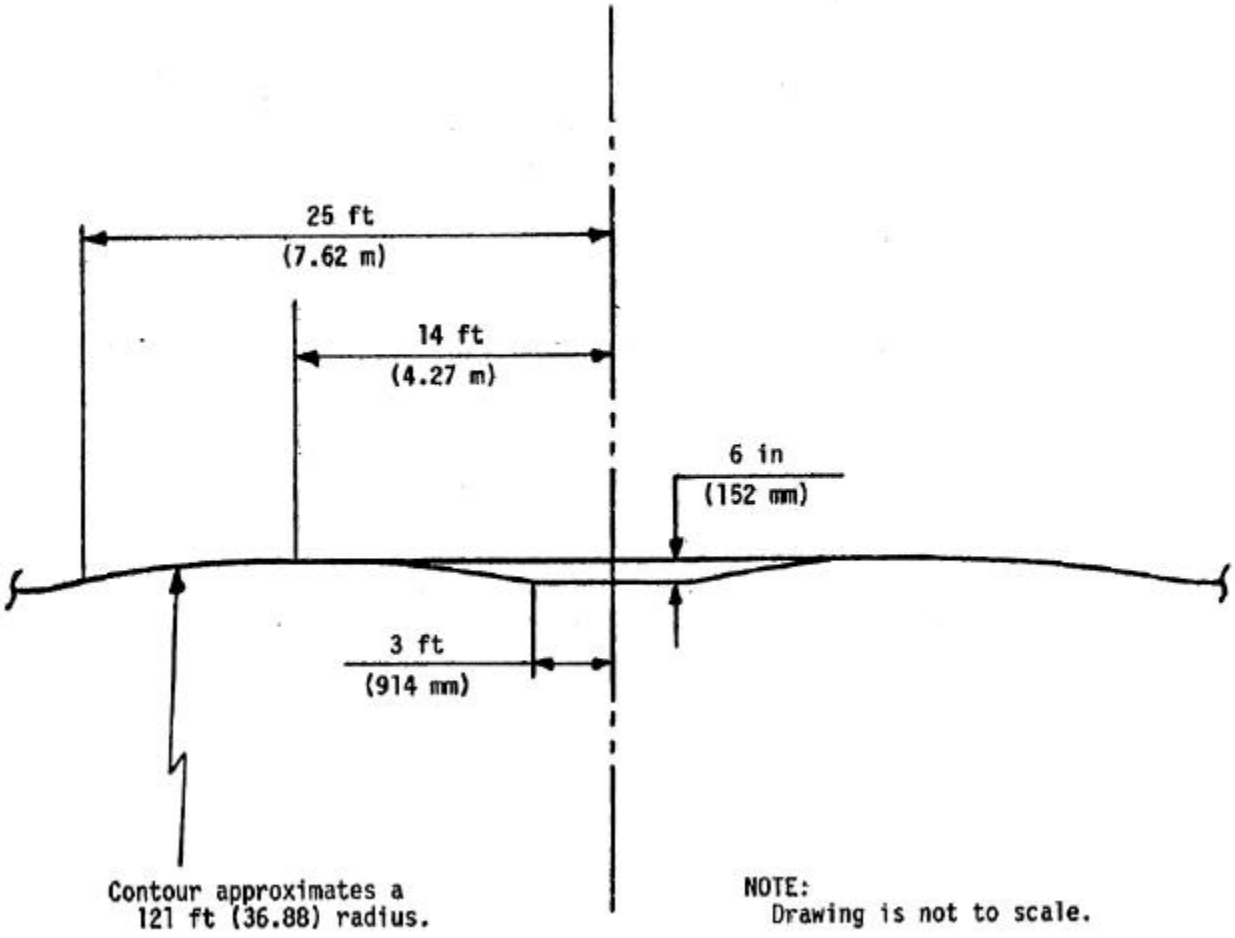
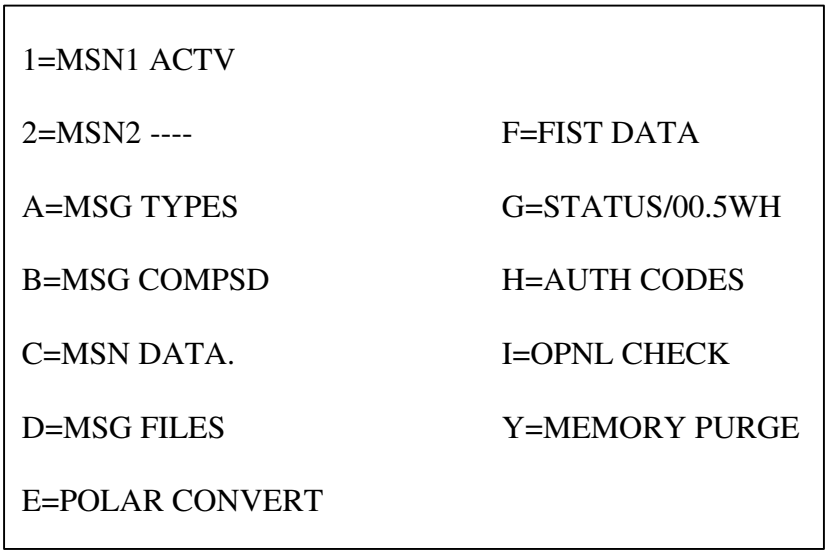


FIGURE 1. Profile of test track (artificial bump).

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NOTE: Battery Status (G=STATUS/00.5WH) may not necessarily read 00.5 WH when system is turned on. Also BATT REQD or BATT WEAK may be displayed. The above display is for the An/PSG-5 DMD only.

FIGURE 2. DMD display.

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